MIL-S-13461C(MU)
14 May 1970
SUPERSEDING
MIL-S-13461B(Ord)
18 September 1962

MILITARY SPECIFICATION

SETTERS, FUZE: M26 AND M28

1. SCOPE

1.1 This specification covers two types of instruments known as Fuze Setter M26 and Fuze Setter M28, which are used in handsetting various types of fuzes.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

D6510486	Fuze, Master MTSQ-M54 and M55A3 for Setter Fuze M26 Instruction Sheet and List of Drawings
F7297021	Fuze, Master MTSQ-M67, M500 and M501 for Setter, Fuze M26
F7647760	Setter, Fuze, M26
F7688626	Setter, Fuze, M28

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F8227024	Fuze, Master MTSQ-T250 for Setter, Fuze M28 - F7688626 Instruction Sheet and List of Drawings
F8227025	Fuze, Master MTSQ-T289 for Setter Fuze, M28 - F7688626 Instruction Sheet and List of Drawings

PACKAGING DATA SHEETS

7647760	Packaging of Setter, Fuze: M26
7688626	Packaging of Setter, Fuze: M28

(Copies of specifications, standards, drawings and packaging data sheets required by contractors in connection with specific procurement functions should be obtained from the procurang activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The Fuze Setters M26 and M28 shall be manufactured in accordance with Drawings F7647760, Setter, Fuze, M26 and F7688626 Setter, Fuze, M28 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Electrical assemblies
(e) Part identification and marking
(f) Workmanship
(g) Soldering electrical connections

3.3 Environmental service conditions.

3.3.1 Extreme operating temperatures.- Unless otherwise specified, the fuze setter shall meet the requirements of 3.5 and 3.6 while exposed to, and thermally stabilized at, ambient temperatures of -40° and +150° Fahrenheit (F).

3.3.2 Vibration.- The fuze setter shall withstand a vibration of 30 ± 5 cycles per second frequency with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes ± 15 seconds. Subsequent to vibration there shall be no evidence of loosened or damaged parts, and the fuze setter shall meet the following requirements of this specification.

3.4 Accuracy.

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3.4.1 M26.- The graduations on the 25-second and 75-second fuze setter scales shall coincide with the scales on the master fuzes within 0.05 and 0.1 second respectively.

3.4.2 M28.- The graduations on the 45-second and 100-second fuze setter scales shall coincide with the scales on master fuzes within 0.05 and 0.1 second respectively.

3.5 Mechanical function.

3.5.1 Thumb screw.- The thumb screw shall remain secure in the tightened position after adjustment.

3.5.2 Handle.- The handle shall be secure in both flat positions as well as in the upright position.

3.6 Illumination.- The lines and figures in the vicinity of the index mark on both the inner and outer scales shall appear clearly defined when the handle is in the upright position and the pushrod is depressed. (Depression of the pushrod will not cause illumination when the handle is in either flat position.)

3.6.1 Light switch.- With a three-volt dc power source applied, the fuze setter shall meet the requirements of 3.6.1.1 and 3.6.1.2.

3.6.1.1 Function.- The switch shall function to complete the circuit when a force of 4 ± 1 pounds is applied to depress the button. The switch shall return to the open position when the force is released.

3.6.1.2 Illumination-stability.- With the switch button depressed, there shall be no flicker or illumination failure when the fuze setter is shaken manually.

3.6.1.3 Continuity.- The light assembly shall demonstrate continuity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his ownfacilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three fuze setters plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III. Examinations and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE I

CLASSIFICATION CRITICAL: NONE MAJOR: AQL 1.0% Defective	REQUIREMENT	TEST PROCEDURE
101. Vibration	3.3.2	4.6.2, 4.6.2.1
102. Scale accuracies, M26	3.4.1	4.6.3
103. Scale accuracies, M28	3.4.2	4.6.3
104. Thumb screw, function	3.5.1	4.6.4
105. Handle, function	3.5.2	4.6.5
106. Continuity	3.6.1.3	4.6.6
107. Illumination	3.6	4.6.7
108. Light switch, function	3.6.1.1	4.6.8
109. Illumination stability	3.6.1.2	4.6.9

MINOR: NONE

NOTE: The tests in Table I shall be conducted at a standard ambient temperature of between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejction" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Three (3) fuze setters shall be selected at random as a special sampling from each 100 produced. The samples shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specifications	3.2	MIL-F-13926 - visual

NOTE: The tests in Table II shall be conducted at a standard ambient temperature of between 60° and 90°F.

4.4.2 Environmental tests.- Three (3) fuze setters shall be selected at random from each 50 produced, or from each month's production whichever occurs first. Each sample shall meet the requirement and test in Table III and shall then meet the requirements for Characteristics 104 through 109, inclusive of Table I.

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TABLE III

NO.	CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
303.	Low temperature operation (-40°F)	3.3.1	4.6.1
304.	High temperature operation (+150°F)	3.3.1	4.6.1

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- When the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Accuracy.- Special testing equipment to test the accuracies of fuze setter M26 (3.4.1) shall conform to fuze, master Drawings D6510486 and F7297021, complete with accessories.

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4.5.2.3.2 Accuracy.- Special testing equipment to test the accuracies of fuze setter M28 (3.4.2) shall conform to fuze, master, Drawings F8227024 and F8227025, complete with accessories.

4.6 Test methods and procedures.

4.6.1 Extreme operating temperatures.- The fuze setters shall be exposed to the temperature conditions specified in 3.3.1, utilizing testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. The fuze setters shall be placed in a cold chamber and the temperature of the chamber reduced gradually -40°F. The fuze setters shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, and while at -40°F the fuze setters shall be subjected to the tests in 4.6.4 to 4.6.8 inclusive to determine compliance with 3.3.1. Upon completion of this test the fuze setter temperature shall be gradually raised to room ambient temperature. The fuze setters shall then be placed into an oven chamber and the temperature shall be raised gradually to +150°F. The fuze setters shall remain at this temperature for a period of 5 hours and at the conclusion of this 5 hour period, and while at +150±F, the fuze setters shall again be subjected to the tests in 4.6.4 to 4.6.8 inclusive to determine compliance with 3.3.1. Upon completion of this test the temperature of the test chamber shall be gradually reduced to room ambient temperature (60° to 90°F) at which time the fuze setters shall be removed from the test chamber. Caution shall be exercised during this test to avoid subjecting the fuze setters to thermal shock.

NOTE: The rate of temperature change in the test chamber shall not exceed 20°F per hour throughout the temperature cycling test of 4.6.1.

4.6.2 Vibration.- Standard vibration equipment provided with means for controlling the direction of vibration, and capable of providing the amplitude and frequency as specified in 3.3.2. The fuze setter rigidly mounted on the vibrating device shall be vibrated along each of the 3 mutually perpendicular axes, relative to its normal operational position. Unless otherwise specified, the test equipment shall be in conformance with the "Test Facilities" of MIL-F-13926.

4.6.2.1 Post vibration.- Subsequent to vibration the fuze setter shall be thoroughly examined for physical defects resulting from poor workmanship and material to determine compliance with 3.3.2 prior to being subjected to the remaining tests. A tactile examination shall be conducted to determine looseness of external parts, or improper functioning of the handle, thumb screw and switch. Fuze setters which have passed the above examinations shall be tested for all the other requirements of this specification.

4.6.3 Accuracy tests.- The accuracy of the fuze setter scales shall be determined utilizing special testing equipment depicted in 4.5.2.3.1 and 4.5.2.3.2 as applicable. To perform test set fuze setter to "S" on

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low scale and secure setting by tightening the thumb screw. Position fuze setter over the nose of the test master as in its intended operation and rotate unit until it is properly seated on the master. Remove the unit under test and visually compare the setting of the master fuze with the unit and it shall coincide within tolerance. Readings shall then be taken with the unit set to four (4) additional equidistant randomly selected settings. The test procedure shall be repeated for the "S" setting and to four (4) additional equidistant randomly selected settings on the high scale of the fuze setter under test. Each pertinent scale of the fuze setter under test shall be visually examined for conformance to the requirements of 3.4 and shall not exceed the tolerance specified therein.

4.6.4 Thumb screw, function.- The thumb screw shall be tested manually for compliance with the requirements of 3.5.1. At any selected setting, when the thumb screw has been tightened, the index shall not move in any direction.

4.6.5 Handle, function.- The handle shall be tested manually for compliance with the requirements of 3.5.2. When the handle is placed in either of the flat positions as well as the upright position, the handle shall remain in such position under normal operating conditions and there shall be no looseness or binding.

4.6.6 Continuity.- A continuity check of the fuze setter shall be made utilizing standard type testing equipment to ascertain conformance with 3.6.1.3.

4.6.7 Illumination.- With the handle in the upright position and the push rod depressed, the illuminated inner and outer scales shall meet the requirements as defined in 3.6 when observed in a darkened area with the unaided eye. With the handle in either of the flat positions and the push rod depressed there shall be no illumination.

4.6.8 Light switch function.- The fuze setter shall be mounted in a suitable position and a commercial type mechanical direct force instrument shall be utilized to apply and measure the force as specified in 3.6.1.1. With the 3 volt dc power source applied, the fuze setter shall then be visually examined in a darkened area to determine compliance with the requirements of 3.6.1.1. When the specified load is applied to depress the light switch button the fuze setter lamp shall light. Removing the load shall return the switch to its normal OFF position, and the light shall be out.

4.6.9 Illumination stability.- Subsequent to the test in 4.6.8, the fuze setter shall be visually examined in the darkened area for the requirements of 3.6.1.2. When manually depressing the switch button, the fuze setter lamp shall light. Hold the depressed button for a period of at least ten seconds. Illumination at this point shall remain steady, and while the setter is being subjected to slight manual shaking, observe the lamp to ascertain no visible evidence of flicker. Releasing the switch

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button shall cause the light to go out and remain out without evidence of flickering. The switch shall be capable of 10 consecutive contacts at 10 second intervals without flicker or illumination failure.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheets 7647760(M26) and 7688626(M28). The level of protection shall be specified in the procurement document.

6. NOTES

6.1 Intended use.- Fuze Setter M26 is intended to be used for handsetting Fuzes TSQ-M77, MT-M67, MTSQ-M54, M55, M500, M501 and M520. Fuze Setter M28 is intended to be used for handsetting Fuzes MTSQ-T250, MTSQ-T289 and Proximity Fuzes M513 and M514.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of an applicable level of preservation, packaging and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Provisions for First Article testing.

Custodian:

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Preparing Activity:

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Project No. 1290-A264

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